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Zeiss Gemini Field Emission Scanning Electron Microscope (FESEM) Replacement

1.0 Scope – This SOW is to provide the requirements for the purchase of a commercially off the shelf (COTS) replacement Field Emission Scanning Electron Microscope (FESEM). 76 MXSS/MXDTAC-1 is requesting the sole source or brand name purchase from Carl Zeiss. The vendor shall provide all hardware, software, accessories, peripherals, documentation, training, shipping, and installation. The vendor will ensure, upon installation, that all detectors and software functions and applications are fully functional along with system components required to operate the FESEM. Any required maintenance will be the sole responsibility of the vendor prior to acceptance.

2.0 Sole Source/Brand Name Justification

The Metallurgical Analysis Section provides critical analyses on jet engine and aircraft components supporting first article testing, production support, prototypes, reverse engineering, failure analysis, and mishap investigations. Parts submitted to the lab are often large or complex geometries that preclude the ability to place in a typical scanning electron microscope chamber. The lab often gets requests to perform surface analyses on production parts or mishap related components that cannot be sectioned or altered in any way. The Zeiss Gemini 460 is the largest chamber available on the market to facilitate this requirement. Avoiding sectioning reduces the lab processing time in half because it eliminates the need for careful preparation, fixturing, cutting, and post sonic agitation cleaning to be able to inspect a part.

Secondly, the lab utilizes a capability known in industry as correlational microscopy where a unique, patented sample holder is used to fix a sample for inspection and that same sample holder can be placed on a Zeiss stereomicroscope or a Zeiss Metallograph. Using the same sample holder with fiduciary marks, digitally marked features are recognized by the Zeiss software and points of interest (coordinates) are shared between each scope. That way a feature of interest such as an NDI indication, crack, pitting, or 10 micron size feature observed on a metallograph can be located at high magnification in the scanning electron microscope, with precision for high magnification and qualitative chemical inspection.

Third, the Oxford system requested is a sub-assembly detector system used for the qualitative chemical analysis and crystallographic mapping on metal alloys. The current system has an Oxford version of this capability with data collected on first articles, mishap and failure analysis investigations that require data to be stored for legal purposes for 7 years should inquiries come up on lab reports in the future. These files are specific to Oxford data formats that cannot be read by other systems and require backwards compatibility (see Section 3.21 for details).

From a budgetary perspective, procurement of another Zeiss translates to significant cost savings from discounts applied to service contract, consumables, and service. The current service contract, FA8126-24-P-0018 has had exceptional performance and will continue to provide support for future service with one current SEM and this new proposed SEM.

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Training: With a staff of 10 engineers, each one has spent upwards of 40 hours (pay rates in the NH-03 pay band) of specialized training on scanning electron microscopy, energy dispersive spectroscopy, and electron backscatter diffraction on the Zeiss and Oxford platforms. Software training and optimization are not a turnkey operation and often require specific applications training from the OEM. To avoid duplication of training costs and efficiency loss, it is most cost-effective to remain with the same platform. A similar comparison would be Solid works or AutoCAD. Many man hours are spent training personnel on using these software packages. Although there are other platforms available, it is the standard format used by design engineers and would be inefficient to have employees adopt another platform.

3.0 Technical Requirements – Key features and requirements of the FESEM are listed below.

3.1 Software

3.1.1 Correlation microscopy capability - Package included with hardware and software to provide this capability. Software includes proven and tested correlative microscopy capability that is compatible with a Zeiss metallograph and Zeiss stereomicroscope with a unique sample holder and is easily recognized by the SEM software to find a point of interest previously identified at a given location by optical methods or vice versa.

3.1.2 Zeiss Zen software that is Windows 10 compliant – Software required to communicate with the three different microscopes in support of the correlational microscopy capability.

3.1.3 Annotations - Capable of making annotations and measurements on screen and saved with image. Annotations can be freely positioned and adjusted. Includes on-screen measurement tools, text and micron bars.

3.1.4 Multiple Image file size and format – Should provide ability to save files to common image types such as TIFF, JPEG, PNG, etc.

3.1.5 Auto Wobbler optimization function – Software can optimize aperture alignment without human wobbler input.

3.1.6 Automatic software switching between detectors and chamber scopes.

3.2 Gun (Electron Beam Source) - Schottky field Emitter type. Accelerating voltage up to 30 KV with a resolution of sub 1 nanometer at 30 KV minimum or better and has in-column beam deceleration.

3.3 Chamber Size - 360mm (X and Y) X 270mm Height Minimum

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3.4 Stage

3.4.1 Stage type - 6 axis eucentric stage with dove tail type specimen holder loading.

3.4.2 Stage Specimen weight capacity - 0.5 kg minimum

3.4.3 Stage Travel (minimum) - X=130mm, Y=130mm, Z=50mm, Tilt: -4 to 70°, Rotation=360° continuous.

3.5 Chamber Pump Down Time - 6 minutes or less

3.6 Chamber door - Drawer or large door type, **Chamber lock (load lock) configurations NOT acceptable.**

3.7 Number of ports – A minimum of fourteen (14).

3.8 Detectors

3.8.1 Standard Secondary electron (SE) detector – For standard Secondary electron imaging.

3.8.2 High resolution 5 diode Backscatter Detector - The 5-diode BSD will support at minimum compositional mode, topographical modes with adjustable polarities on each diode.

3.8.3 In-lens SE - In-lens detector or equivalent for high magnification, high resolution imaging.

3.8.4 Two CCD Chamberscopes – Two (2) chamber cameras to aid in the stage manipulation and sample alignment. One camera to be looking from either front or back while the second observes from the left or right.

3.9 Suspension – Active suspension for vibration dampening. No manual air pumping required.

3.10 Image display - Two high resolution LCD ultrawide monitors 34" minimum

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- 3.11 Un-interruptible Power supply** - 5KVA minimum battery back-up and surge protector with 208 V inputs/outputs and multiple (at least 6) 120V outputs to connect all equipment and detectors running on the SEM. Battery back-up should provide system power for at least 30 minutes to allow for soft shutdowns during power outages.
- 3.12 Apertures** - Multiple aperture size options available on demand via software input. At minimum, must include 20, 30, 60, 120 micron aperture sizes.
- 3.13 Rotary control panel** - Multi-function panel with integrated small sized keyboard (or equivalent), and dedicated rotary knobs for control and adjustment of frequently used SEM parameters.
- 3.14 Joystick Stage Controller** – External stage controller for stage maneuvering.
- 3.15 Integrated Computer** - Intel Xeon processor or equivalent with 32GB RAM running on Microsoft Windows® 10. External Communications Minimum 1 Serial, 2 HDMI outputs minimum, 4 USB 3 ports. Minimum 12 Tb of Hard Drive/SSD, Blue ray DVD optical drive.
- 3.16 Utilities** - Line Power Type 32 A-6 h; 208 V single phase. 50-60 Hz.
- 3.17 System cooling** - Shall be a closed loop system not requiring access to water supplies or drains.
- 3.18 Plasma cleaner for Chamber** – Plasma cleaner required to maintain a clean chamber since the lab works with various materials and post mortem conditions that often have residual volatiles or organic films Cleaner should not take away from the internal available chamber volume.
- 3.19 Technical Support** – On-site technical service, related to commercial warranty coverage, shall be available during business hours of 8:00am – 5:00pm except weekends and federal holidays. Technician shall be on-site within three (3) business days (or as agreed by all parties involved) once determination is made that on-site service is needed. Hours of operation may be changed upon agreement by the contractor, the Contract Officer and the lab personnel.
- 3.20 Spare Filament** – One (1) Spare Thermal Field Emission Filament for replacement.
- 3.21 EDS/EBSD/BEX package** – FESEM to be equipped with an Oxford Brand Energy Dispersive Spectrometer and Electron Backscatter Diffraction Detector system. The current system used in the lab is Oxford name brand and the generated analysis files are unique to Oxford. Analysis files stored in the lab need to be backed up and available for re-analysis for a period of 7 years in support of first articles, mishap investigations, and engineering projects conducted by the lab. Software from other

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vendors would not be able to read the previous files. Oxford analytical package should include the following:

3.21.1 EDS shall be of the most recent new model of 100mm² window size silicon drift detector (SDD) type (no liquid nitrogen) and a windowless model for low kV applications.

3.21.1.1 The following software are required:

3.21.1.1.1 AZtec Live Adv

3.21.1.1.2 Feature

3.21.1.1.3 Large Area Mapping

3.21.1.1.4 Drift Correction

3.21.2 EBSD data shall be capable of providing mapping (local and large area), crystallographic orientation, rotation of crystallographic data, phase analysis, grain size measurements, texture mapping, orientation mapping, IPF, etc. The EBSD detector should be the CMOS FSD type with 5 diodes to ensure rapid acquisition times and able to tilt detector by user.

3.21.2.1 The following software options are required:

3.21.2.1.1 AZtec HKL Adv

3.21.2.1.2 Symmetry S3

3.21.2.1.3 Expanded NIST database of phases to include aerospace alloys for phase identification in alloy systems that are based on aluminum, titanium, iron (steels), nickel (superalloys), copper, magnesium, cobalt.

3.21.3 BEX detector shall be capable of seamlessly combining Backscattered electron and x-ray together.

3.21.3.1 The following software options are required:

3.21.3.1.1 AZtec Live Adv/Expert

3.21.4 AZtec Crystal update to current system for interoperability with new system software so older system can read new files generated.

3.21.5 Dedicated Computer to run EDS and EBSD detectors and software. Must include Intel Xeon Processor with Windows 10 64 bit system. Minimum of 12 TB hard drive/SSD, 32 GB of RAM, Minimum required GPU and graphics card to operate system without lagging.

3.21.6 Two High resolution LCD 34" minimum ultrawide monitors for Oxford system.

3.22 SEM Overall Size - SEM must be able to fit through a 40" doorway.

4.0 Delivery – The contractor shall be responsible for shipping the Field Emission Scanning Electron Microscope to the laboratory. The instrument shall be delivered no later than the agreed-upon contractor's proposed time line without concurrence with the contracting officer

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(CO). The microscope shall be the property of the vendor until it is accepted by the government after all testing is completed and acceptance is granted. Prior to shipping, the vendor shall contact the contract monitor to notify on a delivery date. The supplier is responsible for coordinating with the contract monitor for instrument delivery and escorting to the final location. The supplier is responsible for supplying all the required installation of hardware, software, and fluids (if any). Any matter concerning a change to the scope, prices, terms, or conditions of this contract shall be referred to the Contract Officer (CO).

Only the Government CO can authorize repair actions or changes to the contract. The Contractor shall coordinate any repair actions with the Contract Monitor and CO to obtain approval to proceed. The contractor shall ensure that all repair actions have CO approval before initiating any changes, repair or ordering parts. The contractor shall be responsible for packaging, loading, shipping, unloading, on-site placement and storage at Tinker AFB IAW the manufacturer's commercial practices unless otherwise specified herein. The contractor shall provide a fully authorized representative, on-site, at all times during the receipt, assembly, installation, testing, and acceptance.

- 4.0 Installation** – The government will allocate space for the unit to be installed, including utilities. Utility hook-ups will be within 10 feet of allocated space. The contractor shall connect all electrical connections along with filling all fluids necessary to operate chillers, pumps, etc. The contractor shall have analytical tools installed and functional on the instrument. Calibrations of the instrument shall be performed at the time of installation. All installation procedures shall be completed within thirty (30) calendar days of arrival to the final destination.
- 5.0 Testing** - The tests required for "Quality Assurance Provisions" of this requirement shall be witnessed and approved by a representative of the Metallurgical Analysis Section (76 MXSS/MXDTAC) to confirm instrument performs as advertised, as specified in this SOW. Calibration tools and standards shall be provided by the contractor. The final inspection and acceptance shall be performed by the contractor, and approved/accepted by the contract monitor, no later than thirty (30) days after installation (including the EDS and EBSD analytical tools attached to the FESEM) or as agreed upon by the contractor and Contract Monitor.
- 6.0 Training** – The contractor shall provide on-site training on the operation and use of the new FESEM. EDS/EBSD system provider shall also provide on-site training. **All training sessions** will be performed within three (3) months after installation or as agreed upon by all parties involved. Training shall be at least two days in length to allow sessions for multiple (10) employees. Training will be conducted within business hours of 8:00am – 5:00pm except weekends and federal holidays. Hours of operation may be changed upon agreement by the contractor, the Government CO and the laboratory personnel.
- 7.0 Warranty** – There will be a minimum of one (1) year complete package warranty, as defined in FAR 52.212-4(o), covering parts/labor/software. The commercial warranty will cover the equipment and all of its accessories. The period of the warranty shall begin upon the acceptance of complete and fully operational equipment by the government's Contract Monitor. Any required warranty work to correct defects in material and workmanship, or premature wear, or failure of any machine parts or accessories shall be performed by the contractor at Tinker AFB, OK at no additional cost to the government.
- 8.0 Documentation** - The contractor shall provide a digital copy of a user's manual in the contractor's format and written in English to include:

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- operation,
- trouble-shooting guidelines
- user required maintenance,
- all of the necessary data/specifications documents needed to maintain the accuracy and repeatability of the imaging and measurements by the equipment and its components
- calibration validation procedures, and warranty information.